

Concept 2 | Expected Value and Variance of Multiple Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$E(aX + b) = aE(X) + b$$

Variance

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

Independent

$$\text{Var}(X + Y) = \text{Var } X + \text{Var } Y$$

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Expected Value and Variance of Multiple Variables

Q001 **Written**

Given the probability distributions of X and Y in the given tables, find $E(X+Y)$.

Q002 Written

Three coins are tossed once; let X , Y and Z be the head indicators. Find $E(X+Y+Z)$.

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Expected Value and Variance of Multiple Variables

Q003 **Written**

For the two given tables with $X=1,2,3$ and $Y=4,5,6$, find $E(X+Y)$.

Q004 Written

For the two bags in the given exercise, find the expected total number of red balls drawn.

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Q005 Written

Three independent subsystems each receive two signals with success probability 0.5. Find $E(X+Y+Z)$.

Q006 Written

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z .

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Q007 **Written****Two 500-point coins and two 100-point coins are tossed. Find EZ.**

Q008 Written**A die is thrown twice and the score is $Z=3X+5Y$. Find EZ .**

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Q009 Written

For three 100-point coins and one 50-point coin, find the expected score from heads.

Q010 Written

In two independent raffles, buy two tickets from a $\frac{2}{10}$ winner raffle and two from a $\frac{1}{6}$ winner raffle. Find the expected prizes.

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Q011 Written

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find $E(XY)$.

Q012 Written

For independent X with probabilities $2/5, 2/5, 1/5$ at $1, 2, 3$ and Y uniform on $2, 4, 6$, find $E(XY)$.

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Q013 Written

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find $E(XY)$.

Q014 Written

A card numbered 1 to 4 is replaced and a second card is drawn. Find $E(XY)$.

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Expected Value and Variance of Multiple Variables

Q015 Written

For independent X and Y with $EX=1$ and $EY=2/7$, find $E(XY)$.

Q016 Written

A card X is drawn from 1,2,3 and three-coin heads Y is independent. Find $V(X+Y)$ and $\sigma(X+Y)$.

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Q017 **Written**

Two independent dice are thrown. Find the variance and standard deviation of their sum.

Q018 Written

An independent coin head count X and die score Y are combined. Find $V(X+Y)$ and $\sigma(X+Y)$.

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Expected Value and Variance of Multiple Variables

Q019 Written

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

Q020 Written**Independent X is uniform on 1 to 4 and Y on 1 to 6. Find $V(X+Y)$ and $\sigma(X+Y)$.**

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Expected Value and Variance of Multiple Variables

Q021 Written

Three independent special dice take values 2,4,6 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.

Q022 Written**Three independent dice take values 1,3,5 equally. Find $E(XYZ)$ and $V(X+Y+Z)$.**

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Q023 Written

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

Q024 Written

Three friends have 5, 4 and 3 independent rolls to score a six. Find $E(XYZ)$ and $V(X+Y+Z)$.

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Expected Value and Variance of Multiple Variables

Q025 Written

If X , Y and Z are independent head counts from 4, 3 and 2 fair coins, find $E(XYZ)$.

Q026 MCQ **$E(X+Y)$ equals:****Choose the correct option.****A** $EX-EY$ **B** $EX+EY$ **C** $EXEY$ **D** EX/EY

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Q027 MCQ

For three variables, $E(X+Y+Z)$ equals:**Choose the correct option.**

- A $EXEYEZ$
- B $EX+EY+EZ$
- C $EX-EY-EZ$
- D 0

Q028 MCQ**The expected heads in one fair coin toss is:****Choose the correct option.****A** 0**B** $1/2$ **C** 1**D** 2

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Expected Value and Variance of Multiple Variables

Q029 MCQ

For two signals with $p=0.5$, EX is:**Choose the correct option.**

A 0.5

B 1

C 2

D 4

Q030 MCQ**If $EX=2$ and $EY=5$, $E(X+Y)$ is:****Choose the correct option.****A** 3**B** 7**C** 10**D** 25

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Q031 MCQ

 $E(aX+bY)$ equals:**Choose the correct option.**A $aEX+bEY$ B $abE(XY)$ C $EX+EY$ D $a+b$

Q032 MCQ**The expected score of one fair 100-point coin is:****Choose the correct option.**

- A 25
- B 50
- C 100
- D 200

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Q033 MCQ

For a fair die, $E(3X+5Y)$ with two rolls is:**Choose the correct option.**

- A 14
- B 21
- C 28
- D 35

Q034 MCQ**A fixed entry fee changes:****Choose the correct option.**

- A variance only
- B the mean only
- C neither mean nor variance
- D standard deviation only

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Q035 MCQ

The expected number of winners in 2 of 10 tickets with 2 winners is:**Choose the correct option.**A $1/5$ B $2/5$ C $1/2$ D $4/5$

Q036 MCQ **$E(XY) = EXEY$ requires:****Choose the correct option.**

- A $X = Y$
- B independence
- C zero means
- D equal variances

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Q037 MCQ

For three fair coins, EX is:**Choose the correct option.**A $1/2$

B 1

C $3/2$

D 3

Q038 MCQ**For a fair die, EY is:****Choose the correct option.****A** 2.5**B** 3**C** 3.5**D** 6

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Expected Value and Variance of Multiple Variables

Q039 MCQ

With replacement, two card draws are:**Choose the correct option.**

- A dependent
- B independent
- C identical outcomes
- D impossible

Q040 MCQ**If $EX=2$ and $EY=3$ for independent variables, $E(XY)$ is:****Choose the correct option.****A** 5**B** 6**C** 1**D** 9

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Q041 MCQ

For independent X,Y, $V(X+Y)$ equals:**Choose the correct option.**A $VX-VY$ B $VX+VY$ C $VXVY$

D 0

Q042 MCQ

The variance of a fair die is:**Choose the correct option.**A $7/2$ B $35/12$ C $35/6$ D $1/4$

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Expected Value and Variance of Multiple Variables

Q043 MCQ

Sigma(X+Y) is:**Choose the correct option.**A $V(X+Y)$ B $\sqrt{V(X+Y)}$ C $VX+VY$ D $E(X+Y)$

Q044 MCQ**The variance of the sum of two independent fair dice is:****Choose the correct option.**A $35/12$ B $35/6$ C $70/6$ D $49/4$

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Expected Value and Variance of Multiple Variables

Q045 MCQ

For independent sums, a constant shift changes:**Choose the correct option.**

- A variance
- B standard deviation
- C neither spread measure
- D both spread measures

Q046 MCQ**For independent X, Y, Z , $E(XYZ)$ equals:****Choose the correct option.****A** $EX+EY+EZ$ **B** $EXEYEZ$ **C** 0**D** $E(XY)+EZ$

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Q047 MCQ

For independent X, Y, Z , $V(X+Y+Z)$ equals:**Choose the correct option.**A $VXVYVZ$ B $VX+VY+VZ$ C $VX-VY-VZ$

D 0

Q048 MCQ**For a binomial count with $n=5$ and $p=1/6$, EX is:****Choose the correct option.****A** $1/6$ **B** $5/6$ **C** $5/36$ **D** 6

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Expected Value and Variance of Multiple Variables

Q049 MCQ

For three values 2,4,6 equally likely, EX is:**Choose the correct option.**

A 2

B 3

C 4

D 6

Q050 MCQ**For three independent fair dice, $V(\text{sum})$ is:****Choose the correct option.**A $35/12$ B $35/6$ C $35/4$ D $343/8$

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Expected Value and Variance of Multiple Variables

Quiz WRITTEN 1 **Concept Quiz · Written**

If $EX=4$ and $EY=7$, find $E(3X-2Y)$.

Quiz WRITTEN 2 Concept Quiz · Written

For independent variables with $EX=2$ and $EY=5$, find $E(XY)$.

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Expected Value and Variance of Multiple Variables

Quiz WRITTEN 3 **Concept Quiz · Written**

Two independent fair dice are thrown. Find $V(X+Y)$ and $\sigma(X+Y)$.

Quiz WRITTEN 4 Concept Quiz · Written

Three independent Binomial counts have $(n,p)=(2,1/2),(3,1/2),(4,1/2)$. Find $V(X+Y+Z)$.

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Expected Value and Variance of Multiple Variables

Quiz MCQ 5 [Concept Quiz](#) · [MCQ](#)

If $EX=3$ and $EY=4$, $E(2X+Y)$ is:

Choose the correct option.

A 7

B 10

C 14

D 24

Quiz MCQ 6 Concept Quiz · MCQ

For independent variables $EX=2$, $EY=5$, $E(XY)$ is:

Choose the correct option.

- A 7
- B 10
- C 25
- D $1/2$

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Expected Value and Variance of Multiple Variables

Quiz MCQ 7 [Concept Quiz](#) · MCQ

For two independent die rolls, $V(X+Y)$ is:

Choose the correct option.

A $35/12$

B $35/6$

C $35/4$

D $7/2$

Quiz MCQ 8 Concept Quiz · MCQ

For three independent variables, $V(X+Y+Z)$ is:

Choose the correct option.

- A the product of variances
- B the sum of variances
- C the sum of means
- D always zero

Answer key

Use the question number shown on each page.

Q001 $EX = 35/3, EY = 9/2, \text{so } E(X + Y) = 97/6.$ Q016 $VX=2/3, VY=3/4, \text{so } V(X + Y) = 17/12.$ Q025 $E(XYZ) = 23/21=3.$

Q002 $E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$

and $\sigma = \sqrt{17/12}.$

Q026 B

Q003 $E(X + Y) = EX + EY = 2 + 5 = 7.$

Q017 $V(X + Y) = 35/12 + 35/12 = 35/6;$

Q027 B

Q004 Bag A: $6/7$; Bag B: $6/7$; total $E = 12/7.$

$\sigma = \sqrt{35/6}.$

Q028 B

Q005 Each mean is $2(0.5) = 1, \text{ so}$

Q018 $V = 1/4 + 35/12 = 19/6;$

Q029 B

$E(X + Y + Z) = 3.$

$\sigma = \sqrt{19/6}.$

Q030 B

Q006 $Z = 500X + 100Y, \text{so } EZ = 5001 + 1003/2 = 6500.5$

Q019 $V = 21/4 + 31/4 = 5/4.$

Q031 A

Q007 $EZ = 5001 + 1001 = 600.$

Q020 $V = 5/4 + 35/12 = 25/6;$

Q032 B

Q008 $EZ = 37/2 + 57/2 = 28.$

$\sigma = 5/\sqrt{6}.$

Q033 C

Q009 $EZ = 350 + 25 = 175.$

Q021 $EX=4, VX=8/3; E(XYZ) = 64 \text{ and}$

Q034 B

Q010 $EZ = 22/10 + 21/6 = 11/5.$

$V(\text{sum}) = 8.$

Q035 B

Q011 X and Y are independent, so

Q022 $EX=3, VX=8/3; E(XYZ) = 27 \text{ and}$

Q036 B

$E(XY) = EXEY = 3/27/2 = 21/4.$

$V(\text{sum}) = 8.$

Q037 C

Q012 $EX = 8/5, EY = 4, \text{so } E(XY) = 32/5.$

Q023 $E(\text{product}) = 7/2^3 = 343/8;$

Q038 C

Q013 $E(XY) = EXEY = 3/21 = 3/2.$

$V(\text{sum}) = 335/12 = 35/4.$

Q039 B

Q014 $EX = EY = 5/2, \text{so } E(XY) = 25/4.$

Q024 Means are $5/6, 2/3, 1/2;$

Q040 B

Q015 $E(XY) = EXEY = 2/7.$

$E(\text{product}) = 5/18 \text{ and } V(\text{sum}) = 5/3.$

Q041 B

Q042 B

Q043 B

Q044 B

Q045 C

Q046 B

Q047 B

Q048 B

Q049 C

Q050 C

Quiz WRITTEN 1 -2

Quiz WRITTEN 2 10

Quiz WRITTEN 3 $V = 35/6, \sigma = \sqrt{35/6}$

Quiz WRITTEN 4 $9/4$

Quiz MCQ 5 B

Quiz MCQ 6 B

Quiz MCQ 7 B

Quiz MCQ 8 B